



PROJECT		JOB NO.				
CONTRACTOR :						
SUB-CONTRACTOR : Winner Engineering Pte Ltd						
TRADE	: ACMV SERVICES		DRAWING NO :			
FUNCTIONAL- TESTING COMMISSIONING CHECKLIST (AHU/FCU - VRF UNIT / VRV UNIT)						
Unit Designation :						
Unit Make :						
Model no.						
Location :						
Serial no :						
S/N	Section	Inspection / Test item	CHECKED CONDITION (Please tick inside checkbox)			Remarks
			Yes	No	NA	
1	General	Check that the equipment is clean, free of debris, and undamaged before commencing operations.				
2	Documents	Verify chiller tag, make, model, serial number, refrigerant and rated capacity.				
3	Electrical	Record supply voltage and applicable phase sequence / balance.				
4	Startup	Perform manufacturer commissioning/test-run procedure and record fault history.				
5	Fan & motor	Verify indoor/outdoor fan operation, rotation and available speed settings.				
6	Electrical	Record unit/compressor/fan operating current as accessible under the approved test procedure.				
7	Mechanical	Check noise, vibration, mountings and casing during operation.				
8	Controls	Test local/remote Start, Stop, Auto and provided controller functions.				
9	Controls	Test setpoint changes and sensor readings against calibrated instruments.				
10	Controls	Verify thermostat demand, compressor modulation/cycling and restart delay.				
11	Cooling	Record outdoor ambient, room/return air and supply/off-coil temperatures at stable cooling duty.				
12	Refrigerant	Record available service-tool pressures, temperatures and operating data as required by the manufacturer.				
13	Heating	Test heating, changeover and applicable defrost function if a heat-pump unit.				
14	Drainage	Check drainage during operation at required fan speeds and cooling duty.				
15	Drainage	Test condensate pump and high-level alarm/shutdown where fitted.				
16	Safety	Test applicable safety inputs and fault indications by approved simulation/service procedure.				
17	Controls	Test power-restoration behaviour and retained settings using an approved controlled test.				
18	VRF system	Verify communication, indoor-unit count, addresses and zone/controller mapping.				
19	VRF system	Test each indoor unit individually and the system under concurrent demand.				

20	VRF system	Verify expansion-valve and branch-selector response using manufacturer diagnostic procedure.				
21	VRF system	Test mode priority, heat-recovery operation and simultaneous heating/cooling if supported.				
22	DX AHU	Test fan enable, airflow proof and DX cooling/heating interlocks where an AHU is connected.				
23	DX AHU	Test specified coil-temperature protection and AHU interface response.				
24	Airside	Test motorized dampers, NRDs and VAV interfaces where fitted.				
25	Airflow	Measure total airflow, outlet airflow and AHU/ducted-unit external static pressure.				
26	Airflow	Measure fresh-air quantity and filter pressure drop where specified.				
27	Environment	Verify room temperature and RH performance where specified.				
28	BMS	Verify central/room controller and BMS commands, status, alarms, schedules and sensor values.				
29	Interlocks	Test fire/smoke shutdown and associated dampers where required.				
30	Refrigerant safety	Test installed refrigerant detection, alarms and mitigation controls where required.				
31	Completion	Record final setpoints, operating mode, schedules and controller settings.				

Comments to follow up Action :

Yes = satisfactory; No = rectify and re-inspect; NA = not applicable (state reason). Blank = not checked.

CHECKED BY:		WITNESSED BY	INSPECTED BY
NAME			
DESIGNATION :			
COMPANY :	Winner Engineering Pte Ltd		
SIGNATURE :			
DATE :			